

HIAS: Hybrid Intelligence Approach for Soil Classification and Recommendation of Crops

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ABSTRACT ORIGINAL

Agriculture is the largest industry in the world, and it is vital to any country's economic development. Farmers' failure to select the appropriate crop for cultivation is a significant and severe mistake that results in poor agricultural output. For the surplus of crop production, a soil classification and crop recommendation system were developed, which used a semantically driven hybrid intelligence method. In this paper, the Hybrid Intelligence Approach for Soil (HIAS) model was presented and tested with several baseline models using southern South Australian Soil datasets to evaluate the proper prediction. Besides, ontology is generated from the summarised content of agricultural and geological eBooks. Textbooks and then the query and Soil dataset are pre-processed. The semantic similarity is calculated using Jaccard similarity, Cosine similarity, and SemantoSim measure under the Squirrel search algorithm to ensure that the relevant result is optimised and set entities formed from a set of recommendations are optimised. The term enrichment is done using MediaWiki and WikiData, and then the LSTM classifier is used for classification. Finally, the precision, accuracy, recall, F-measure, and FNR for the soil classification and crop recommendation system are evaluated. The precision percentage of 99.78% and the lowest FNR of 0.01 is obtained from the HIAS model. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.